Description

Image



Caption

These boat fenders illustrate that PVC is tough, weather resistant and easy to form and color.

The material

PVC - Vinyl - is one of the cheapest, most versatile and - with polyethylene - the most widely used of polymers and epitomizes their multi-facetted character. In its pure form - as a thermoplastic, tpPVC - it is rigid, and not very tough; its low price makes it a cost-effective engineering plastic where extremes of service are not encountered. Incorporating plasticizers creates flexible PVC, elPVC, a material with leather-like or rubber-like properties, and used a substitute for both. By contrast, reinforcement with glass fibers gives a material that is sufficiently stiff, strong and tough to be used for roofs, flooring and building panels. Both rigid and flexible PVC can be foamed to give lightweight structural panels, and upholstery for cars and domestic use. Blending with other polymers extends the range of properties further: vinyl gramophone records were made of a vinyl chloride/acetate co-polymer; blow molded bottles and film are a vinyl chloride/acrylic copolymer.

Compositional summary

(CH2CHCl)n

General properties

Density	1.29e3	-	1.46e3	kg/m^3
Price	* 1.03	-	1.41	GBP/kg
Date first used	1940			

Mechanical properties

Young's modulus	2.2	-	3.1	GPa
Shear modulus	* 0.75	-	1.1	GPa
Bulk modulus	* 3.49	-	5.44	GPa
Poisson's ratio	* 0.395	-	0.405	
Yield strength (elastic limit)	37.6	-	45.5	MPa

Tensile strength		38	-	46	MPa
Compressive strength	*	37	-	44.3	MPa
Elongation		40	-	80	% strain
Hardness - Vickers		12			HV
Fatigue strength at 10^7 cycles	*	16.6	-	21.1	MPa
Fracture toughness	*	3.63	-	3.85	MPa.m^0.5
Mechanical loss coefficient (tan delta)	*	0.00966	-	0.0166	

Thermal properties

Glass temperature		79.9	-	87.9	°C
Maximum service temperature	*	49.9	-	62.9	°C
Minimum service temperature		-18.2	-	7.85	°C
Thermal conductor or insulator?		Good insulator			
Thermal conductivity		0.147	-	0.209	W/m.°C
Specific heat capacity		1e3	-	1.1e3	J/kg.°C
Thermal expansion coefficient		65	-	81	µstrain/°C

Electrical properties

Electrical conductor or insulator?		Good insulator			
Electrical resistivity		1e20	-	1e22	µohm.cm
Dielectric constant (relative permittivity)		3	-	3.2	
Dissipation factor (dielectric loss tangent)		0.02	-	0.03	
Dielectric strength (dielectric breakdown)		13.8	-	19.7	MV/m

Optical properties

Transparency		Transparent			
Refractive index		1.53	-	1.54	

Critical Materials Risk

High critical material risk?		No			

Processability

Castability		1	-	2	
Moldability		4	-	5	
Machinability		3	-	4	
Weldability		5			

Durability: water and aqueous solutions

Water (fresh)		Excellent			
Water (salt)		Excellent			

Soils, acidic (peat)	Excellent
Soils, alkaline (clay)	Excellent
Wine	Excellent

Durability: acids

Acetic acid (10%)	Excellent
Acetic acid (glacial)	Excellent
Citric acid (10%)	Excellent
Hydrochloric acid (10%)	Excellent
Hydrochloric acid (36%)	Excellent
Hydrofluoric acid (40%)	Excellent
Nitric acid (10%)	Excellent
Nitric acid (70%)	Excellent
Phosphoric acid (10%)	Excellent
Phosphoric acid (85%)	Excellent
Sulfuric acid (10%)	Excellent
Sulfuric acid (70%)	Excellent

Durability: alkalis

Sodium hydroxide (10%)	Excellent
Sodium hydroxide (60%)	Excellent

Durability: fuels, oils and solvents

Amyl acetate	Unacceptable
Benzene	Unacceptable
Carbon tetrachloride	Limited use
Chloroform	Unacceptable
Crude oil	Acceptable
Diesel oil	Excellent
Lubricating oil	Excellent
Paraffin oil (kerosene)	Excellent
Petrol (gasoline)	Excellent
Silicone fluids	Acceptable
Toluene	Unacceptable
Turpentine	Excellent
Vegetable oils (general)	Excellent
White spirit	Acceptable

Durability: alcohols, aldehydes, ketones

Acetaldehyde	Limited use
Acetone	Unacceptable
Ethyl alcohol (ethanol)	Excellent
Ethylene glycol	Excellent
Formaldehyde (40%)	Excellent
Glycerol	Excellent
Methyl alcohol (methanol)	Excellent

Durability: halogens and gases

Chlorine gas (dry)	Unacceptable
Fluorine (gas)	Unacceptable
O2 (oxygen gas)	Limited use
Sulfur dioxide (gas)	Excellent

Durability: built environments

Industrial atmosphere	Excellent
Rural atmosphere	Excellent
Marine atmosphere	Excellent
UV radiation (sunlight)	Fair

Durability: flammability

Flammability	Self-extinguishing

Durability: thermal environments

Tolerance to cryogenic temperatures	Unacceptable
Tolerance up to 150 C (302 F)	Acceptable
Tolerance up to 250 C (482 F)	Unacceptable
Tolerance up to 450 C (842 F)	Unacceptable
Tolerance up to 850 C (1562 F)	Unacceptable
Tolerance above 850 C (1562 F)	Unacceptable

Geo-economic data for principal component

Annual world production, principal component	4.3e7			tonne/yr
Reserves, principal component	6.56e8	-	7.25e8	tonne

Primary material production: energy, climate change and water

Climate change (CO2-eq) (virgin grade production)	* 2.91	-	3.21	kg/kg
Embodied energy (virgin grade production)	* 65.5	-	72.2	MJ/kg
Water usage	* 181	-	200	l/kg

Material processing: energy

Polymer extrusion, embodied energy	* 7.96	-	8.8	MJ/kg
Polymer molding (injection, blow), embodied energy	* 18.2	-	20.6	MJ/kg
Machining, coarse, embodied energy	* 16.3	-	18.1	MJ/kg
Machining, fine, embodied energy	* 19.6	-	22	MJ/kg
Machining, grinding, embodied energy	* 23	-	25.9	MJ/kg

Material processing: climate change

Polymer extrusion, climate change (CO2-eq)	* 0.522	-	0.578	kg/kg
Polymer molding (injection, blow), climate change (CO2-eq)	* 1.22	-	1.4	kg/kg
Machining, coarse, climate change (CO2-eq)	* 1.06	-	1.18	kg/kg
Machining, fine, climate change (CO2-eq)	* 1.28	-	1.44	kg/kg
Machining, grinding, climate change (CO2-eq)	* 1.5	-	1.7	kg/kg

Recycling and end of life

Recycle	✓			
Functional recycle	✗			
Climate change (CO2-eq), recycling	* 0.96	-	1.06	kg/kg
Embodied energy, recycling	* 12.4	-	13.7	MJ/kg
Recycle fraction in current supply	1			%
Combust for energy recovery	✓			
Heat of combustion (net)	* 17.5	-	18.4	MJ/kg
Combustion CO2	* 1.37	-	1.44	kg/kg
Landfill	✓			
Biodegrade	✗			
Toxicity rating	Non-toxic			
A renewable resource?	✗			

Environmental notes

The vinyl chloride monomer is thoroughly nasty stuff, leading to pressure to discontinue production. But properly controlled, the processing is safe, and the polymer PVC has no known harmful effects. Disposal, however, can be a problem: thermal degradation releases chlorine, HCl and other toxic compounds, requiring special high-temperature incineration for safety.

Recycle mark



Supporting information

Design guidelines

In its pure form, PVC is heavy, stiff and brittle. Plasticizers can transform it from a rigid material to one that is almost as elastic and soft as rubber. Plasticized PVC is used as a cheap substitute for leather, which it can be made to resemble in color and texture. It is less transparent than PMMA or PC, but it also costs much less, so it is widely used for transparent, disposable containers. PVC is available as film, sheet or tube. It can be joined with polyester, epoxy or polyurethane adhesives. It has excellent resistance to acids and bases and good barrier properties to atmospheric gasses, but poor resistance to some solvents.

Technical notes

PVC can be a thermoplastic or a thermoset. There are many types of PVC: expanded rigid PVC, type I, type II, CPVC, acrylic/PVC blend, clear PVC.

Typical uses

tpPVC: pipes, fittings, profiles, road signs, cosmetic packaging, canoes, garden hoses, vinyl flooring, windows and cladding, vinyl records, dolls, medical tubes. elPVC: artificial leather, wire insulation, film, sheet, fabric, car upholstery.

Tradenames

Conoco, Dural, Ethyl, Flexalloy, Geon, Hy-vin, Keysor, Locovyl, Novatemp, Oxyclear, Polyvin, Satinflex, Sicron, Solvic, Solvin, Superkleen, Trosiplast, Unichem, Vestolit, Vinoflex, Vistel

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ProcessUniverse

Producers

Reference

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